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Three-Schema Architecture, Data Independence and DBMS Architectures

Understanding the Layers Behind Modern Database Systems



Learning Outcomes

- *By the end of our session, we will be able to:*
- * Explain why data abstraction is required.
- * Understand the Three-Schema Architecture.
- * Differentiate between external, conceptual, and internal schemas.
- * Explain logical and physical data independence.
- * Understand schema mappings.
- * Compare centralized, client-server, and distributed database architectures.
- * Briefly relate these concepts to modern cloud database systems.



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Why Do We Need Data Abstraction?

- *Problem*

- A real-world DB is extremely complex.
 - Example: University Database Contains
 - * Students
 - * Faculty
 - * Courses
 - * Departments
 - * Attendance
 - * Library
 - * Hostel
 - * Fees
 - * Examination

- Definition

- Data Abstraction is the process of hiding unnecessary implementation details while exposing only the essential information.

- Different users require different views.

- *Example:*
 - A Student wants
 - * Marks
 - * Attendance
 - A Teacher wants
 - * Student List
 - * Grades
 - The Accounts Office wants
 - * Fee Details
 - Each user should see only the required data.



Levels of Data Abstraction

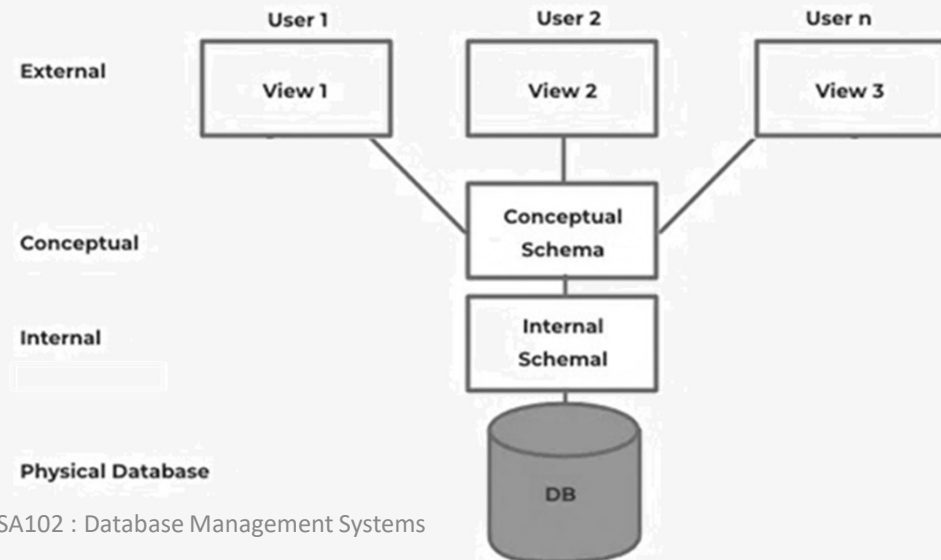
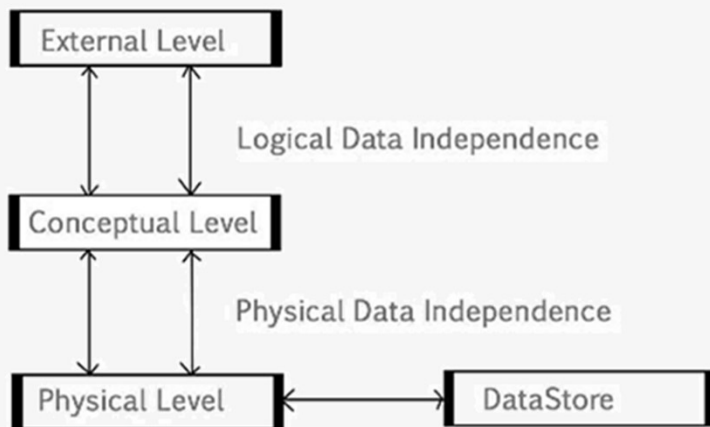
- Database systems hide complexity through three levels of abstraction.
 - 1. Physical Level
 - Describes
 - * Storage
 - * Indexes
 - * File organization
 - * Memory allocation
 - Concerned with how data is stored.
 - 2. Logical Level
 - Describes
 - * Tables
 - * Attributes
 - * Relationships
 - * Constraints
 - Concerned with what data is stored .
 - 3. View Level
 - Shows only the information required by a user.
 - Concerned with what the user sees.
- *Example*
 - Student Portal
 - Shows
 - * Name
 - * Marks
 - but hides
 - * Salary of Faculty
 - * Internal Database Structure



ANSI/SPARC Three-Schema Architecture

- Definition
- The Three-Schema Architecture is a framework proposed by ANSI/SPARC to separate user applications from the physical database.
- *It consists of*
 1. External Schema
 2. Conceptual Schema
 3. Internal Schema

- *Purpose*
 - * Data abstraction
 - * Data independence
 - * Security
 - * Multiple user views
 - * Easier maintenance



CSA102 : Database Management Systems



External Schema (View Level)



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- Definition
- The External Schema defines how individual users view the database.
- Each user may have a different view.
- *Characteristics*
- * User-specific
- * Simplified
- * Secure
- * Hides unnecessary information

- *Example*
- Student View
- | Roll | Name | Marks |
- Teacher View
- | Roll | Name | Marks | Attendance |
- Accounts View
- | Roll | Fee Status |
- Each view is an External Schema.
- *Advantages*
- * Improved security
- * Simplicity
- * Customized access

Conceptual Schema (Logical Level)

- Definition
- The Conceptual Schema represents the complete logical structure of the entire database.
- *It includes*
 - * Tables
 - * Relationships
 - * Constraints
 - * Keys
- *Characteristics*
 - * Entire database
 - * Independent of storage
 - * Common view for the organization
- *Example*
 - University Database
 - Student
 - Faculty
 - Course
 - Department
 - Enrollment
 - All relationships exist here.



Internal Schema (Physical Level)

- Definition
- The Internal Schema describes how the data is physically stored.

- *It specifies*
- * Storage structures
- * File organization
- * Record format
- * Indexes
- * Compression
- * Storage locations

Example

Student table stored as

- * B+ Tree Index
- * Heap File
- * SSD Storage

Applications do not see these details.

Importance: Improves

- * Performance
- * Storage utilization
- * Retrieval speed



Schema Mapping

- The three schemas are connected through mappings.
- External-Conceptual Mapping
- Converts
 - User View→Conceptual Database
- Conceptual-Internal Mapping
- Converts
 - Logical Design→Physical Storage
- Why are mappings important?
 - Because they allow changes at one level without affecting the others.



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Data Independence

- Definition
- Data Independence is the ability to modify one level of the database without affecting higher levels.
- It is one of the greatest advantages of DBMS.
- *Benefits*
- * Easier maintenance
- * Lower development cost
- * Better scalability
- * Flexible database evolution



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Physical Data Independence

- Definition
- The ability to change the internal schema without affecting the conceptual schema.
- *Changes Allowed*
 - * New indexes
 - * File organization
 - * Compression
 - * Storage devices
 - * Partitioning
- *Example*
 - Old storage → Hard Disk
 - New storage → SSD
- Applications continue working without modification.
- *Another Example*
 - Adding a B+ Tree index to speed up queries.
 - No application changes are needed.



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Logical Data Independence

- Definition
- The ability to modify the conceptual schema without affecting external schemas.
- Possible Changes
 - * Add a new table
 - * Add new attributes
 - * Split a table
 - * Merge tables
- *Example*
 - Initially: Student (Roll, Name)
 - Later: Student (Roll, Name, Email)
 - Student Portal continues working without changes if mappings are maintained.
- *Note:* Logical Data Independence is more difficult to achieve than Physical Data Independence.



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Physical vs Logical Data Independence

• Physical Data Independence	Logical Data Independence
• -----	-----
• Internal level changes	Conceptual level changes
• Easier to achieve	Harder to achieve
• Storage changes	Structure changes
• Example: Adding Index	Example: Adding New Table
• Memory Trick	
• Physical = Storage Changes	
• Logical = Structure Changes	



Centralized Database Architecture

- Definition
- In a Centralized Database System, the entire database resides on a single server.
- *Characteristics*
 - * One database server
 - * Multiple users
 - * Central administration
- *Advantages*
 - * Easy management
 - * Better consistency
 - * Easier backup
- *Disadvantages*
 - * Single point of failure
 - * Limited scalability
- *Example*
- Small college database server.



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Client-Server Database Architecture

- Definition
- In Client-Server Architecture, the client requests services while the database server processes requests.
- Components
- *Client*
 - * User Interface
 - * Application
- *Server*
 - * DBMS
 - * Database
- *Working*
 - Client sends SQL query → Server processes → Result returned
- *Examples*
 - * Banking systems
 - * ERP
 - * Hospital Management System
 - * Online Examination Systems
- *Advantages*
 - * Better performance
 - * Resource sharing
 - * Centralized security



Modern Database Architectures (Overview)

- Today's systems often extend the client-server model.
- Distributed Databases
 - Data stored across multiple locations.
 - Example
 - Global Banking Network
- Cloud Databases
 - Managed over the Internet.
 - Examples
 - * Amazon RDS
 - * Google Cloud SQL
 - * Microsoft Azure SQL Database

- Benefits
 - * High availability
 - * Elastic scalability
 - * Automated backup
- NoSQL Systems (Brief Pointer)
- Large-scale applications such as
 - * Netflix
 - * Amazon
 - * Facebook
- often combine SQL and NoSQL databases depending on workload.



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Summary

- Today we learned
- ✓ Data Abstraction
- ✓ Three Levels of Abstraction
- ✓ ANSI/SPARC Three-Schema Architecture
- ✓ External Schema
- ✓ Conceptual Schema
- ✓ Internal Schema
- ✓ Schema Mappings
- ✓ Data Independence
- ✓ Physical Data Independence
- ✓ Logical Data Independence
- ✓ Centralized Architecture
- ✓ Client-Server Architecture
- ✓ Modern Distributed & Cloud Database Concepts



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Learning Reflections

- 1. Why is data abstraction necessary in DBMS?
- 2. Explain the Three-Schema Architecture with a neat diagram.
- 3. Differentiate between the external, conceptual, and internal schemas.
- 4. Explain logical and physical data independence with suitable Examples.
- 5. Why is logical data independence more difficult to achieve than physical data independence?
- 6. Compare centralized and client-server architectures.
- 7. How do cloud databases build upon traditional client-server architecture?



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Homework

- Explain the ANSI/SPARC Three-Schema Architecture with suitable examples. Discuss the concepts of data abstraction, schema mapping, and physical and logical data independence. Also briefly compare centralized, client-server, and modern distributed/cloud database architectures.





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Thanks!

📌 Heads Up!

These slides are just a **guide** — not the full story!

To really learn, make sure to read your book, and explore other resources too.